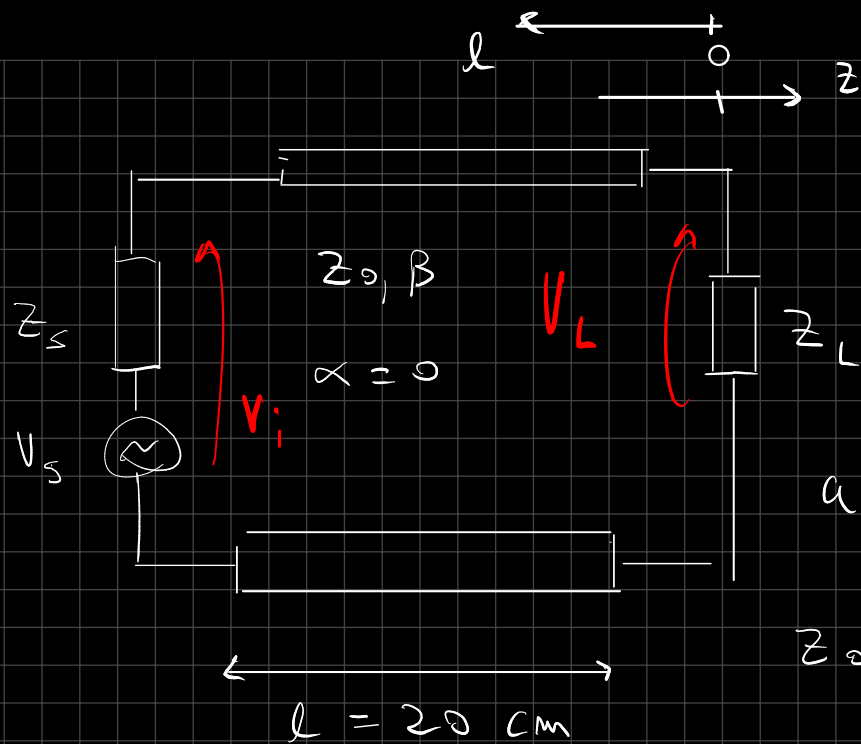




$$V_s = 1 \text{ V}$$

$$Z_s = 50 \Omega$$

$$Z_L = 1 \text{ k}$$

$$f = 16 \text{ MHz}$$

$$L = 5 \text{ nH/cm}$$

$$C = 0,4 \text{ pF/cm}$$

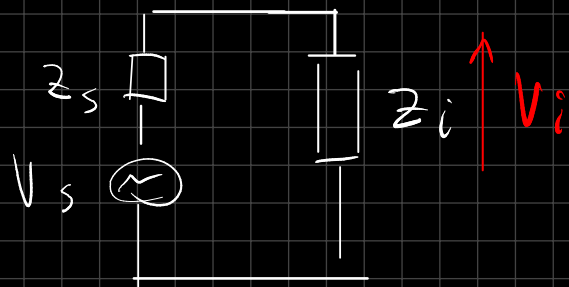
$$R = G = 0$$

a) γ, Z_0

$$Z_0 = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{L/C} = 111,8 \Omega$$

$$\gamma = 0 + j\beta = j\omega \sqrt{L \cdot C} = j0,281 \text{ rad/cm}$$

b) $V_i?$ $V_L?$



$$V_i = \frac{Z_i}{Z_i + Z_s} \cdot V_s$$

Sim pérdidas

$$z = -l$$

$$z_i = z(z) \stackrel{!}{=} z_0 \frac{z_L - j z_0 \tan(\beta \cdot z)}{z_0 - j z_L \tan(\beta \cdot z)} \stackrel{!}{=} z_0 \frac{z_L + j z_0 \tan(\beta \cdot l)}{z_0 + j z_L \tan(\beta \cdot l)} =$$

$$= 32,3 + j 138,5$$

$$V_i = 0,7028 e^{j 18^\circ}$$

$$j V_L? \rightarrow V(z) = V_0^+ e^{-j \beta \cdot z} + V_0^- e^{j \beta \cdot z} = V_0^+ e^{j \beta l} + V_0^- e^{-j \beta l}$$

$$\Gamma_L = \frac{V_0^-}{V_0^+} = \frac{z_L - z_0}{z_L + z_0} = 0,7989$$

$$\rightarrow V(l) = V_0^+ (e^{j \beta \cdot l} + \Gamma_L e^{-j \beta l})$$

$$V(l=0) = V_L$$

$$V(l=l) = V_i \rightarrow \text{Le començem}$$

predesacat V_0^+

$$V_o^+ = V_i \cdot \frac{e^{-j\beta \cdot l}}{1 + \Gamma_L e^{-j2\beta l}} = 0,5728 \text{ V} + j 0,2383 \text{ V}$$

$$V(l=0) = V_L = V_o^+ (1 + \Gamma_L) = 1,12 \text{ V} e^{j22,8^\circ}$$

→ mayor q' la tensión q' entrega el generador. Interferencia constructiva

$$c) \langle P_i \rangle \quad \langle P_L \rangle$$

$$\langle P_i \rangle = \frac{1}{2} \operatorname{Re} \{ V_i \cdot I_i^* \} = \begin{cases} \text{si } P > 0 \rightarrow \text{potencia hacia la carga} \\ \text{si } P < 0 \rightarrow \text{" q' se refleja} \end{cases}$$

$$= \frac{1}{2} \operatorname{Re} \left\{ V_i \cdot \left(\frac{-V_i}{Z_i} \right)^* \right\} = \frac{1}{2} |V_i|^2 \operatorname{Re} \left\{ \frac{1}{|Z_i|} e^{-j\theta_{Z_i}} \right\}$$

$$= \frac{|V_L|^2}{2|Z_L|} \cos(\phi_{Z_L}) = 623 \mu W$$

$$\langle P_L \rangle = \frac{1}{2} \operatorname{Re} \left\{ V_L \cdot \left(\frac{V_L}{Z_L} \right)^* \right\} = \frac{|V_L|^2}{2|Z_L|} \cdot \cos(\phi_{Z_L}) = \frac{|V_L|^2}{2|Z_L|} = 623 \mu W = \langle P_i \rangle$$

$\langle P_L \rangle = \langle P_i \rangle \rightarrow$ En la línea no se disipa nada

Ahora te la complico: $R = 5 \Omega/cm$

Si $R \ll \omega L \rightarrow$ bajas pérdidas

$Z = R + jX \rightarrow$ reactancia
 $\hookrightarrow$ resistencia

Si $G \ll \omega C \rightarrow$ bajas pérdidas

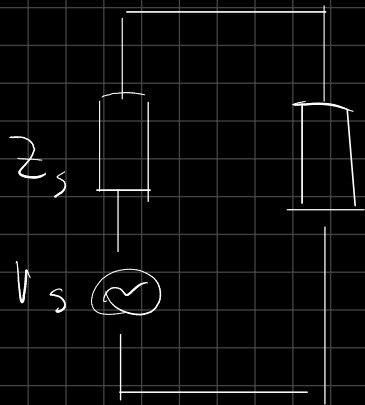
$Y = G + jS \rightarrow$ susceptancia
 $\hookrightarrow$ conductancia

$\omega L = 31.4 \Omega/cm \rightarrow$ No es $\gg R$

No VALE Aproximar

$$Z_0 = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R + j\omega L}{j\omega C}} = (112,2 - j8,87) \Omega$$

$$\gamma = \sqrt{Z \cdot Y} = \sqrt{(R + j\omega L)(j\omega C)} = \underbrace{2,23 \frac{\text{Np}}{\text{m}}}_{\alpha} + j \underbrace{28,2 \text{ rad/m}}_{\beta}$$



$$Z_i = Z_0 \cdot \frac{Z_L + Z_0 \tanh(\gamma \cdot l)}{Z_0 + Z_L \tanh(\gamma \cdot l)}$$

→ Esta es la única cuenta q' cambia
El resto es igual

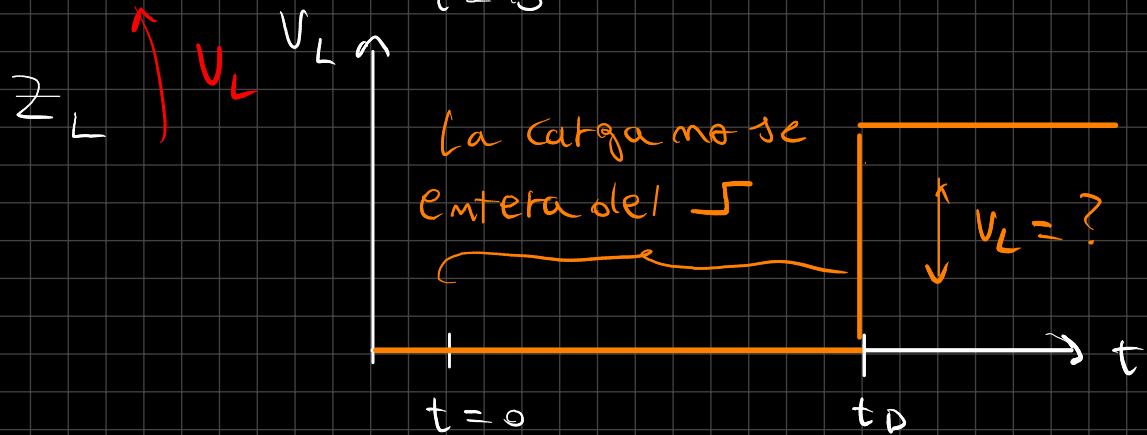
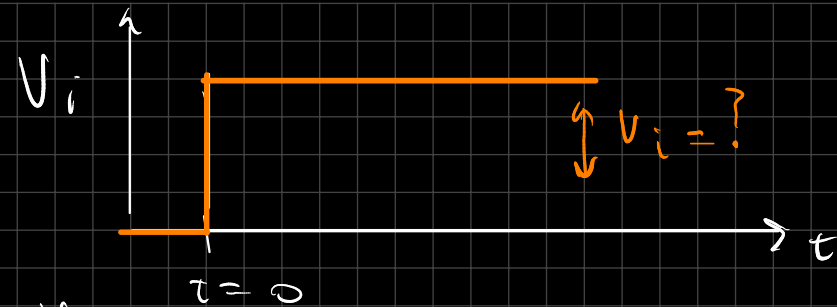
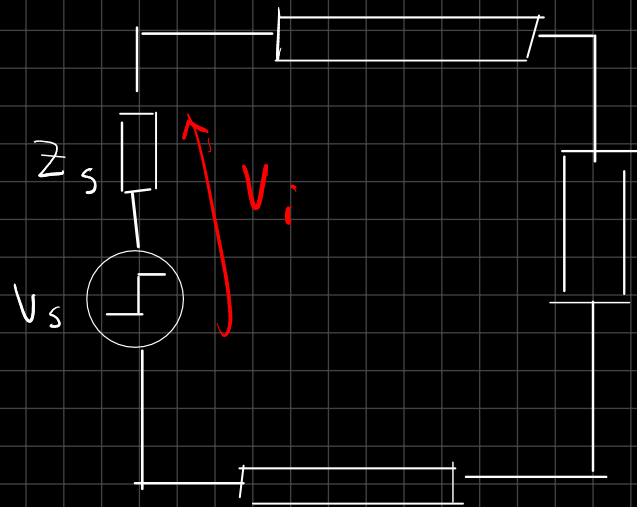
$$V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$$

$$V_L = V(z=0) = V_0^+ (1 + \Gamma_L)$$

$$\langle P_i \rangle = 1,82 \text{ mW} \quad \langle P_L \rangle = 0,289 \text{ mW}$$

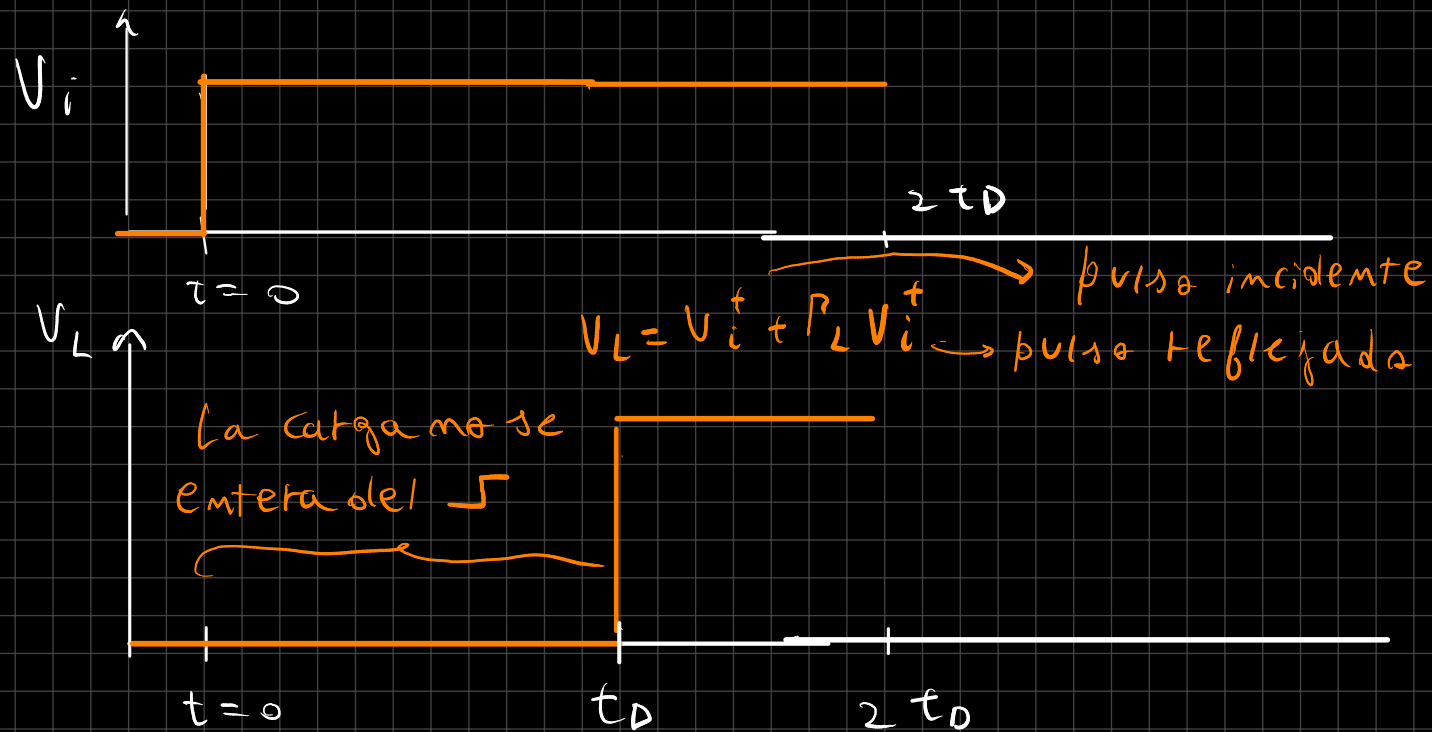
$$\langle P_o \rangle = \langle P_i \rangle - \langle P_L \rangle = 1,53 \text{ mW}$$

En otro orden de cosas:



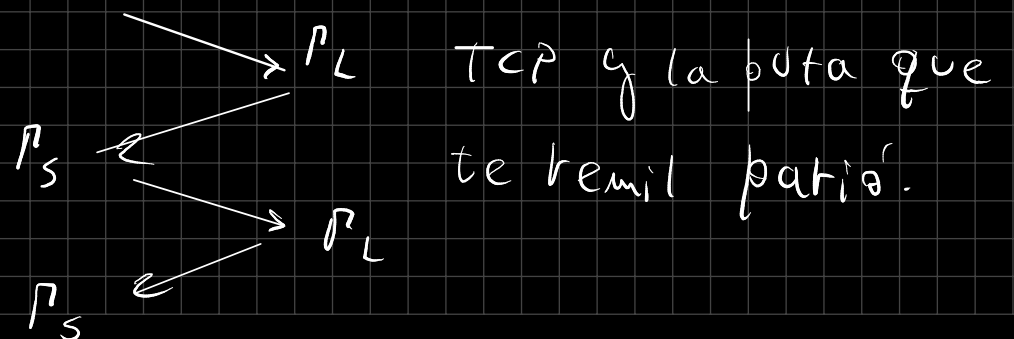
Antes: $V_i = \frac{Z_i}{Z_s + Z_i} \cdot V_s$ → ¿Cuánto vale Z_i si todavía no está circulando corriente por la carga?

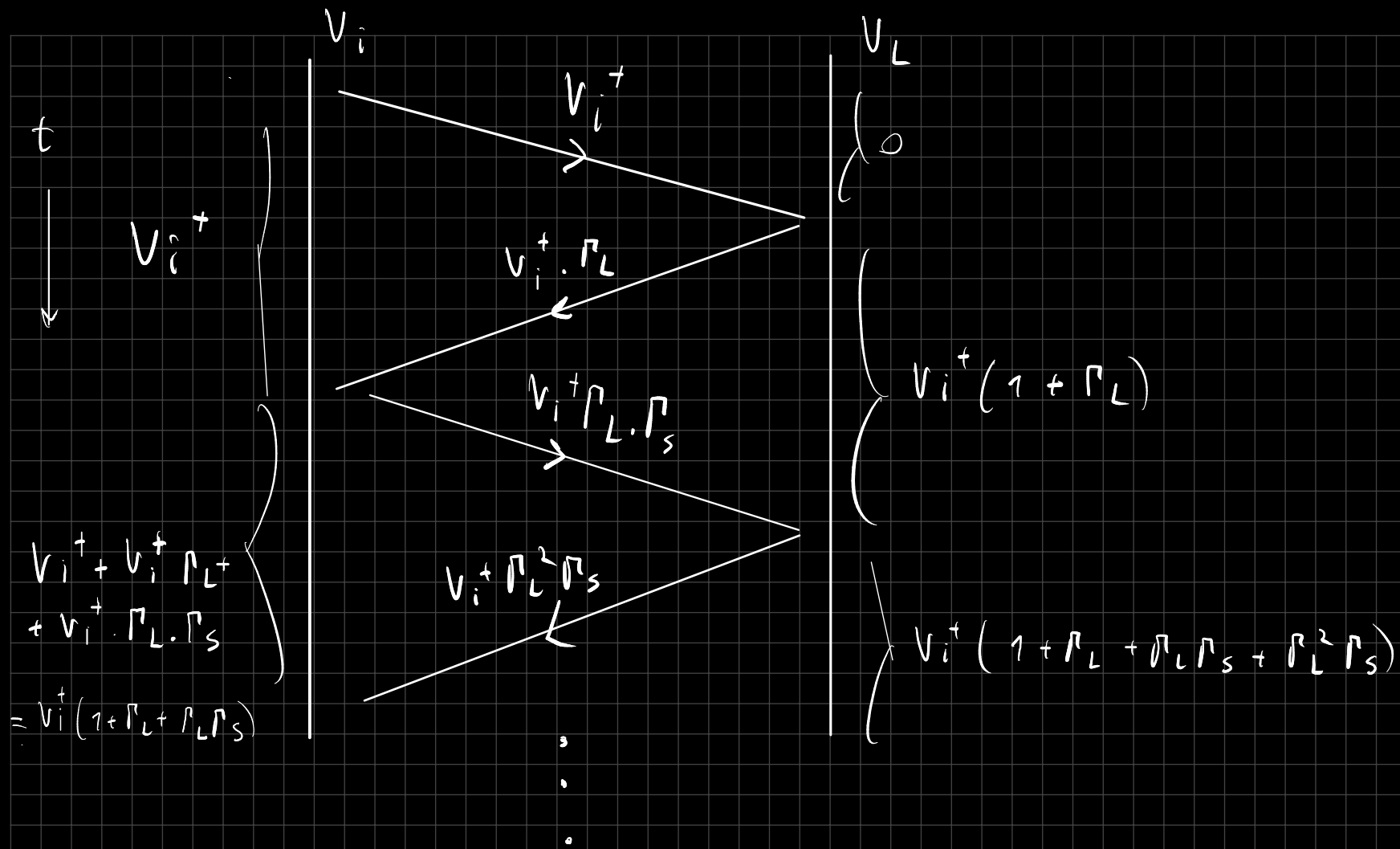
Ahora: $V_i^+ = \frac{Z_o}{Z_s + Z_o} \cdot V_s$ → ahora esta es la relación entre V e I q' ve el pulso



El pulso vuelve y rebota otra vez en el generador

$$\Gamma_s = \frac{Z_s - Z_0}{Z_s + Z_0}$$





$$V_L \xrightarrow{t \rightarrow \infty} V_i^+ \left(1 + \Gamma_L + \Gamma_L \Gamma_S + \Gamma_L^2 \Gamma_S + \Gamma_L^2 \Gamma_S^2 + \Gamma_L^3 \Gamma_S^2 + \dots \right) =$$

$$= V_i^+ \left[(1 + \Gamma_L \Gamma_S + \Gamma_L^2 \Gamma_S^2 + \dots) + (\Gamma_L + \Gamma_L^2 \Gamma_S + \Gamma_L^3 \Gamma_S^2 + \dots) \right] =$$

$$= V_i^+ \left[(1 + \Gamma_L \Gamma_S + \Gamma_L^2 \Gamma_S^2 + \dots) + \Gamma_L (1 + \Gamma_L \Gamma_S + \Gamma_L^2 \Gamma_S^2 + \dots) \right] =$$

$$= V_i^+ (1 + \Gamma_L) \cdot \sum_{n=0}^{+\infty} (\Gamma_L \Gamma_S)^n = V_i^+ (1 + \Gamma_L) \frac{1}{1 - \Gamma_L \Gamma_S}$$

$\downarrow$
 $|\Gamma_L| < 1$

$V_L \xrightarrow{t \rightarrow \infty} \frac{1 + \Gamma_L}{1 - \Gamma_L \Gamma_S} \cdot V_i^+ = \frac{Z_L}{Z_S + Z_L} \cdot V_S$

→ Como si fuera un cable!

$\downarrow$
 ¿Desarrollar Γ_L y Γ_S ?